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ANNUAL EVALUATION 2024-2025
MATHEMATICS

E-903

Class : 9

Time : $2\frac{1}{2}$ hours

Score : 80

Instructions

- There is a 'cool off' time of 15 minutes in addition to the writing time. Use this time to get familiar with questions and plan your answers.
- Read the instructions carefully before answering the questions.
- Keep in mind, the score and time while answering the questions. Give explanations wherever necessary.
- No need to simplify irrationals like $\sqrt{2}, \sqrt{3}, \pi$ etc., using approximations unless you are asked to do so.

Answer any 3 Questions from 1 to 4. Each question carries 2 scores. **(3 x 2 = 6)**

1. The marks of 8 students in a maths test are given below.

   70, 65, 45, 54, 72, 66, 58, 50

   What is the mean mark of the students ?

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2. The sum of two numbers is 21 and their difference is 5.
   What are the numbers ?

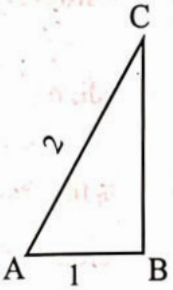
3. The area of a circle is 4π square centimetres.
   What is its radius ?

4. In the given figure,
   what part of the total area of the circle is unshaded region ?



Answer any 4 Questions from 5 to 10. Each question carries 3 scores. **(4 x 3 = 12)**

5. In the figure, ABC is a right triangle and
   AB = 1 metre, AC = 2 metres.
   a) What is the length of BC in metres ?
      (1, $\sqrt{2}$, $\sqrt{3}$, 2)
   b) Calculate the perimeter of the square having side BC and correct it upto a centimetre.



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6. The base edge of a regular hexagonal prism is 8 centimetres and its height is 15 centimetres.

   a) Find the base area of the prism.

   b) Calculate the volume of the prism.

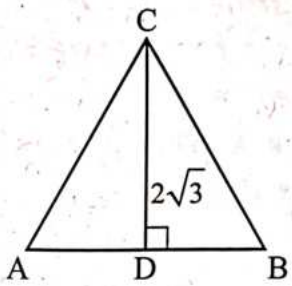
7. In the figure, ABC is an equilateral triangle.

   CD = $2\sqrt{3}$ centimetres

   a) What is the length of AB in centimetres ?

   ($2, 2\sqrt{3}, 4, 4\sqrt{3}$)

   b) Find the area of triangle ABC.



8. In a circle, the length of an arc of central angle 80^0 is 12 centimetres.

   a) What is the length of an arc of central angle 20^0 in this circle?

   b) What is the length of an arc of central angle 20^0 in a circle with radius one and half times of this circle?

9. Draw an equilateral triangle of perimeter 13 centimetres.

10. The relation between radius and perimeter of circles is given in the following table.

Radius (Centimetre)	Perimeter (Centimetre)
1	2π
2	4π
3	6π
r	.....

   a) Complete the table.

   b) Is the circumference of the circle proportional to the radius ?
   If so, what is the proportionality constant ?

Answer any 8 Questions from 11 to 21. Each question carries 4 scores. **(8 x 4 = 32)**

11. The base radii of two cylinders are in the ratio 1 : 2 and their heights are in the ratio 4 : 3.

   a) What is the ratio of their lateral surface areas ?

   b) If the lateral surface area of the first cylinder is 30 square centimetres, what is the lateral surface area of the second cylinder ?

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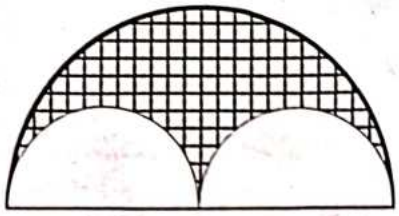
12. In a polynomial $p(x) = x^2 + 3x + 2$,

a) $p(0) =$ _____

b) $p(1) =$ _____

c) Find $\frac{p(1)+p(-1)}{p(0)}$.

13. In the figure, the diameter of large semicircle is 8 centimetres. Two semicircles with half its diameter are drawn.



a) What is the radius of the small semicircle ?

b) Find the area of the large semicircle.

c) What is the area of the shaded region ?

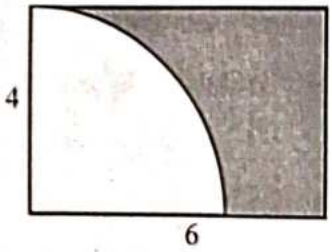
14. Find x satisfying each of the equations below.

a) $|x+2| = 5$

b) $|x-6| = |x+2|$

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15. In the figure, the sides of rectangle are 6 centimetres and 4 centimetres. A sector is drawn with a vertex as centre.



a) What is the area of the rectangle ?

b) Find the area of the sector.

c) Find the area of the shaded region.

16. The base edges of a triangular prism are 5 centimetres, 12 centimetres and 13 centimetres. Height of the prism is 20 centimetres.

a) What is the base perimeter of the prism ?

b) Find the lateral surface area of the prism.

c) What is the total surface area of the prism. ?

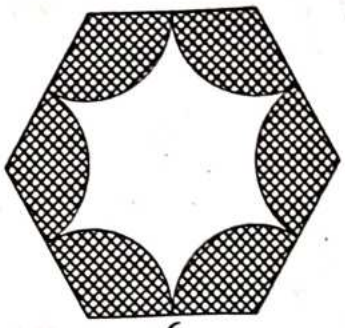
17. If the sides of a rectangle are decreased by 1 metre, its area would be 99 square metres. If increased by 1 metre, it would be 143 square metres.

a) What is the area of the rectangle ?

b) Find the length and breadth of the rectangle.

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18. In the figure, one side of a regular hexagon is 6 centimetres. Sectors are drawn with vertices of this hexagon as centre.



a) What is the central angle of a sector ?
(60^0, 90^0, 120^0, 180^0)

b) What is the perimeter of the regular hexagon ?

c) Find the perimeter of the shaded region.

19. The length and breadth of rectangles of area 100 square centimetres are given in the table.

Length (Centimetre)	Breadth (Centimetre)
50	$\frac{100}{50} = 2$
20	5
10	.....
4	.....

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a) Complete the table.

b) Using the table, write the relation as an algebraic equation by taking length as x centimetre and breadth as y centimetre.

c) How do we state it in terms of proportion ?

20. Draw the graph of the polynomial $p(x) = x - 2$.

21. The table below shows members in a farmers' club grouped according to their ages.

Age	Number of members
40	8
45	6
47	10
50	7
55	4
58	5

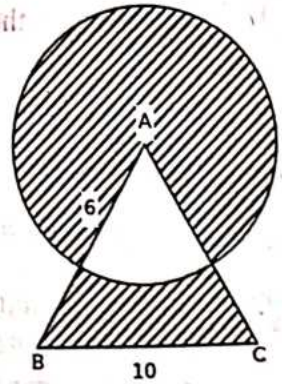
a) How many members are there in the club ?

b) Find the mean age of the members of the club.

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Answer any 6 Questions from 22 to 29. Each question carries 5 scores. **(6 x 5 = 30)**

22. In the figure, a circle is drawn with centre as one vertex of an equilateral triangle of side 10 centimetres. Radius of the circle is 6 centimetres.



a) $\angle BAC =$ _____
($30^0, 45^0, 60^0, 72^0$)

b) Find the area of the unshaded sector.

c) Find the total area of the shaded region.

23. A block of wax is in the shape of a square prism of base edge 12 centimetres and height 25 centimetres. A cylinder of maximum size is carved out from it.

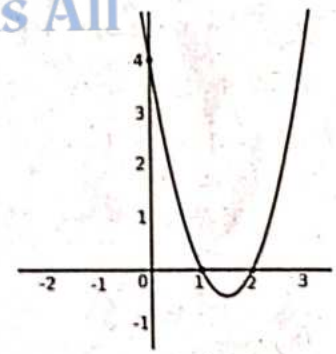
a) What is the radius of the cylinder?

b) Find the volume of the cylinder.

c) How many identical cubes of sides 1 centimetre can be made by using the remaining wax ?

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24. The graph of a second degree polynomial is given.



a) $p(1) = 0$, $p(0) =$_____, $p(2) =$_____

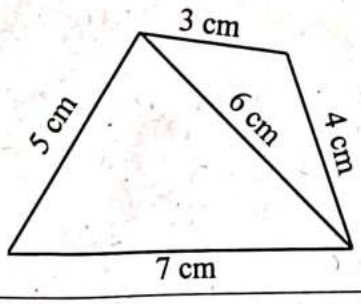
b) Find the polynomial.

25. In an auditorium, there are 15 pillars in the shape of square prism. The base edge of the pillar is 40 centimetres and height is 4 metres.

a) What is the lateral surface area of one pillar?

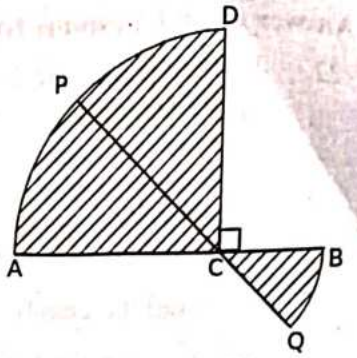
b) What is the total cost of painting all the pillars at rupees 90 per square metre ?

26. See the picture of the given quadrilateral. Draw the quadrilateral with the same angles as this and sides scaled by $1\frac{1}{2}$ times.



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27. In the figure, AB=12 centimetres, AC : CB = 2 : 1.
PQ is the bisector of $\angle ACD$.



a) What is the measure of $\angle BCQ$?

b) AC = _____

c) CB = _____

d) Find the area of the shaded region.

28. The consumption of electricity of 20 households in a locality is given in the following table.

Electricity consumption (Unit)	Number of households
100 - 150	4
150 - 200	2
200 - 250	5
250 - 300	6
300 - 350	3

Find the mean consumption of electricity.

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29. Read the following mathemetical pattern and answer the questions given below.

$\left(1+\frac{1}{2}\right)\div\left(1-\frac{1}{2}\right)=3$

$\left(\frac{1}{2}+\frac{1}{3}\right)\div\left(\frac{1}{2}-\frac{1}{3}\right)=5$

$\left(\frac{1}{3}+\frac{1}{4}\right)\div\left(\frac{1}{3}-\frac{1}{4}\right)=7$

.....

.....

.....

a) Write the next line of this pattern.

b) $\left(\frac{1}{15}+\frac{1}{16}\right)\div\left(\frac{1}{15}-\frac{1}{16}\right)=$ _____

c) Write the 25th line.

d) $\left(\frac{1}{29}+P\right)\div\left(\frac{1}{29}-P\right)=59$, find P.

e) $\left(\frac{1}{n}+\frac{1}{n+1}\right)\div\left(\frac{1}{n}-\frac{1}{n+1}\right)=$ _____